

Revised Generation Interconnection System Impact Study Report

For Queue Project Q16-0607

South Lake Geneva 138 kV
October 28, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0607. The Interconnection Customer proposes a 104.2 MW battery energy storage facility interconnecting at the South Lake Geneva 138 kV substation in the Pennsylvania area. The request was received on April 14, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 255 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 18 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$63,170,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Central Beach – Paxon 138 KV	Westwood Generation Llc – Coronet 500 KV line	259.5	194	38.4
Fitzpatrick – Central Beach 138 KV	Westwood Generation Llc – Coronet 500 KV L	240.2	216	38.4
Fitzpatrick – Talenenergy Martins Crk Llc Harrisburg 138 KV line	Coronet – Westwood Generation Llc 500 KV Line	228.0	241	38.4
Dumbarton Tap – Paxton Crk Cogeneration 138 KV Line	Talenenergy Martins Crk Llc Harrisburg – Little Chute 345 KV	206.7	219	14.9
Ayer – Paxton Creek Cogeneration 138 KV line	Little Chute – Talenenergy Martins Crk Llc Harrisburg 345 KV Line	192.0	228	14.9
Ayer – Talenenergy Martins Crk Llc Harrisburg 138 KV	Little Chute – Talenenergy Martins Cree Llc Harrisburg 345 KV	168.8	244	14.9
Macarthur – Fort Gage 138 KV	Talenenergy Martins Creek Llc Harrisburg – Little Chute Substation 345 KV	164.9	270	10.4
Dumbarton Tap – Macarthur 138 KV	Little Chute – Talenenergy Martins Creek Llc Harrisburg 345 KV	164.5	260	10.4
Little Chute – Hunt 138 KV	Little Chute – Talenenergy Martins Crk Llc Harrisburg 345 KV line	152.0	251	7.0
Hunt – Ft Gage 138 KV Line	Talenenergy Martins Crk Llc Harrisburg – Little Chute 345 KV	152.0	241	7.0
SouthLake Geneva – Paxon 138 KV	Talenenergy Martins Crk Llc Harrisburg 500 kV/345 kV Transformer	151.2	206	58.3

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Elk City Llc 138 kV/1 kV Transformer Bank	Talenenergy Martins Crk Llc Harrisburg – Little Chute 345 KV	138.6	437	7.6
Paxon Substation – L L Wilkins 138 KV L	Wetwood Generation Llc – Coronet 500 KV	134.9	169	21.0
Elk City Llc 345 kV/1 kV Transformer	Little Chute – Talenenergy Martins Creek Llc Harrisburg 345 KV L	124.4	487	7.6
Hummelstown – S Lake Geneva 138 KV	Talenenergy Martins Creek Llc Harrisburg 500 kV/345 kV Transformer	114.7	230	41.7
L L Wilkins – Wolverine Creek 138 KV Line	Westwood Generation Llc – Coront 500 KV	109.1	197	21.0
Little Chute 345/138 kV Transformer	Little Chute – Talenenergy Martins Crk Llc Harrisburg 345 KV	109.0	382	6.0
Hummelstown – River Bend 138 KV	Talenenergy Martins Creek Llc Harrisburg 500/345 kV Auto	105.3	239	41.7

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Central Beach – Paxon 138 KV	Rebuild line with higher-capacity conductor	\$1,977,000	80% (\$1,585,000)
Fitzpatrick – Central Beach 138 KV	Rebuild line with higher-capacity conductor	\$1,296,000	80% (\$1,039,000)
Fitzpatrick – Talenenergy Martins Crk Llc Harrisburg 138 KV line	Rebuild line with higher-capacity conductor	\$2,754,000	80% (\$2,208,000)
Dumbarton Tap – Paxton Crk Cogeneration 138 KV Line	Rebuild line with higher-capacity conductor	\$2,837,000	100% (\$2,837,000)
Ayer – Paxton Creek Cogeneration 138 KV line	Rebuild line with higher-capacity conductor	\$3,249,000	100% (\$3,249,000)
Ayer – Talenenergy Martins Crk Llc Harrisburg 138 KV	Rebuild line with higher-capacity conductor	\$410,000	100% (\$410,000)
Macarthur – Fort Gage 138 KV	Rebuild line with higher-capacity conductor	\$1,273,000	100% (\$1,273,000)
Dumbarton Tap – Macarthur 138 KV line	Rebuild line with higher-capacity conductor	\$1,494,000	100% (\$1,494,000)
Little Chute – Hunt 138 KV	Rebuild line with higher-capacity conductor	\$2,345,000	100% (\$2,345,000)
Hunt – Ft Gage 138 KV Line	Rebuild line with higher-capacity conductor	\$2,736,000	100% (\$2,736,000)
Paxon – S Lake Geneva 138 KV	Rebuild line with higher-capacity conductor	\$1,296,000	100% (\$1,296,000)
Elk City Llc 138/1 kV Transformer Bank	Replace transformer with higher-rated unit	\$4,489,000	100% (\$4,489,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Paxon Substation – L L Wilkins 138 KV L	Rebuild line with higher-capacity conductor	\$1,506,000	100% (\$1,506,000)
Elk City Llc 345 kV/1 kV Transformer	Replace transformer with higher-rated unit	\$15,361,000	100% (\$15,361,000)
Hummelstown – S Lake Geneva 138 KV	Rebuild line with higher-capacity conductor	\$5,559,000	100% (\$5,559,000)
L L Wilkins – Wolverin Creek 138 KV	Reconductor line	\$777,000	100% (\$777,000)
Little Chute 345/138 kV Transformer	Replace transformer with higher-rated unit	\$14,268,000	100% (\$14,268,000)
Hummelstown – River Bend 138 KV	Reconductor line	\$738,000	100% (\$738,000)
	Total allocated to project		\$63,170,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.